

Universidad de Costa Rica

Escuela de Ciencias de la Computacion e Informatica

CI-0123 Proyecto Integrador de Sistemas Operativos y Redes de Comunicación de Comunicación de datos

Conexión L2 con L3

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1.Configuracion del L3

```
!
version 12.2
no service pad
service timestamps debug uptime
service timestamps log uptime
no service password-encryption
!
hostname L3-0403
!
!
no aaa new-model
ip subnet-zero
ip routing
!
!
!
!
no file verify auto
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
interface FastEthernet0/1
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 230,240
 switchport mode trunk
!
interface FastEthernet0/2
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 230,240
 switchport mode trunk
!
interface FastEthernet0/3
!
```

```
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
```

```
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 no ip address
!
interface Vlan230
 ip address 172.16.123.62 255.255.255.240
!
interface Vlan240
 ip address 172.16.123.78 255.255.255.240
!
ip classless
ip http server
!
!
control-plane
!
!
line con 0
line vty 5 15
!
!
end
```

2.Configuracion L2 ISLA 4

```
Building configuration...

Current configuration : 3288 bytes
!
version 12.2
```

```
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Pris
!
boot-start-marker
boot-end-marker
!
!
!
!
no aaa new-model
system mtu routing 1500
ip dhcp excluded-address 172.16.123.65 172.16.123.66
!
ip dhcp pool ISLA4
   network 172.16.123.64 255.255.255.240
   default-router 172.16.123.65
!
!
no ip domain-lookup
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
!
!
interface FastEthernet0/1
```

```
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/2
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/3
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/4
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/5
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/6
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/7
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/8
 switchport access vlan 240
 switchport mode access
```

```
 spanning-tree portfast
!
interface FastEthernet0/9
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/10
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/11
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/12
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/13
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/14
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/15
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
```

```
interface FastEthernet0/16
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/17
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/18
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/19
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/20
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/21
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/22
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/23
 switchport access vlan 240
```

```
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
 switchport trunk allowed vlan 240
 switchport mode trunk
!
interface GigabitEthernet0/2
!
interface Vlan1
 no ip address
!
interface Vlan240
 ip address 172.16.123.66 255.255.255.240
!
ip default-gateway 172.16.123.65
ip http server
ip http secure-server
vstack
!
line con 0
line vty 5 15
!
end
```

3.Configuración L2 ISLA 3

```
!
version 12.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname ISLA3
!
```

```
boot-start-marker
boot-end-marker
!
!
!
!
no aaa new-model
system mtu routing 1500
ip dhcp excluded-address 172.16.123.49 172.16.123.54
!
ip dhcp pool ISLA3
  network 172.16.123.48 255.255.255.240
 --More--          default-router 172.16.123.49
 --More--        !
 --More--        !
 --More--        no ip domain-lookup
 --More--        !
 --More--        !
 --More--        !
 --More--        !
 --More--        !
 --More--        !
 --More--        spanning-tree mode pvst
 --More--        spanning-tree extend system-id
 --More--        !
 --More--        vlan internal allocation policy ascending
 --More--        !
 --More--        !
 --More--        !
 --More--        interface FastEthernet0/1
 --More--         switchport access vlan 230
 --More--         switchport mode access
 --More--         spanning-tree portfast
 --More--        !
 --More--        interface FastEthernet0/2
 --More--         switchport access vlan 230
 --More--         switchport mode access
```

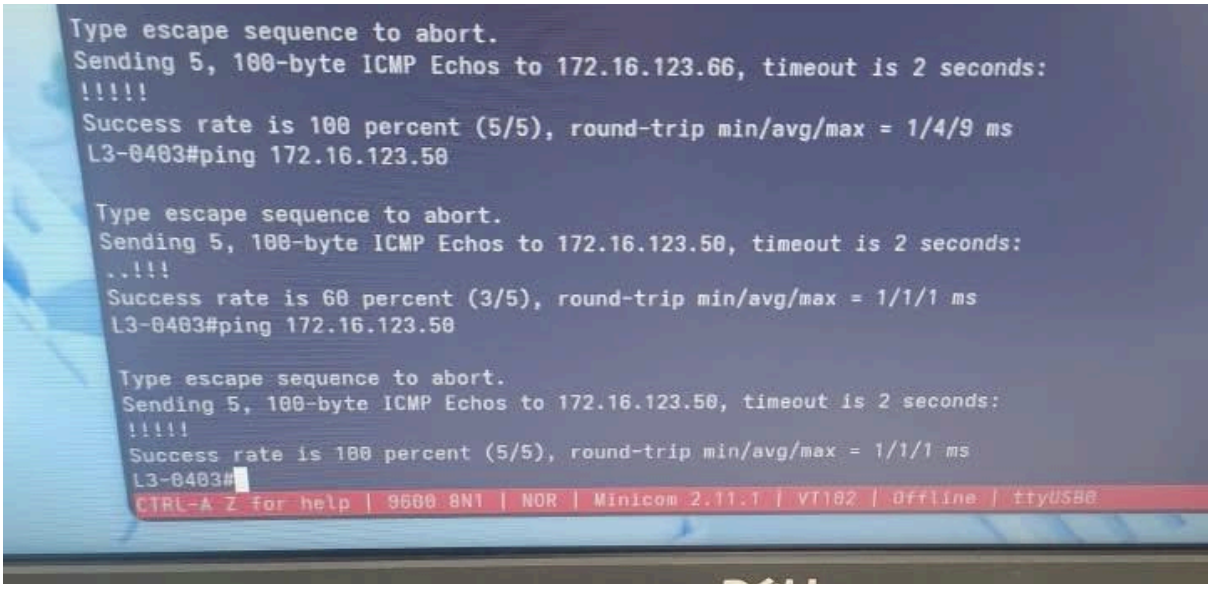
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/3
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/4
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/5
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/6
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/7
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/8
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/9
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !

--More-- interface FastEthernet0/10
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/11
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/12
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/13
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/14
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/15
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/16
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/17
--More-- switchport access vlan 230

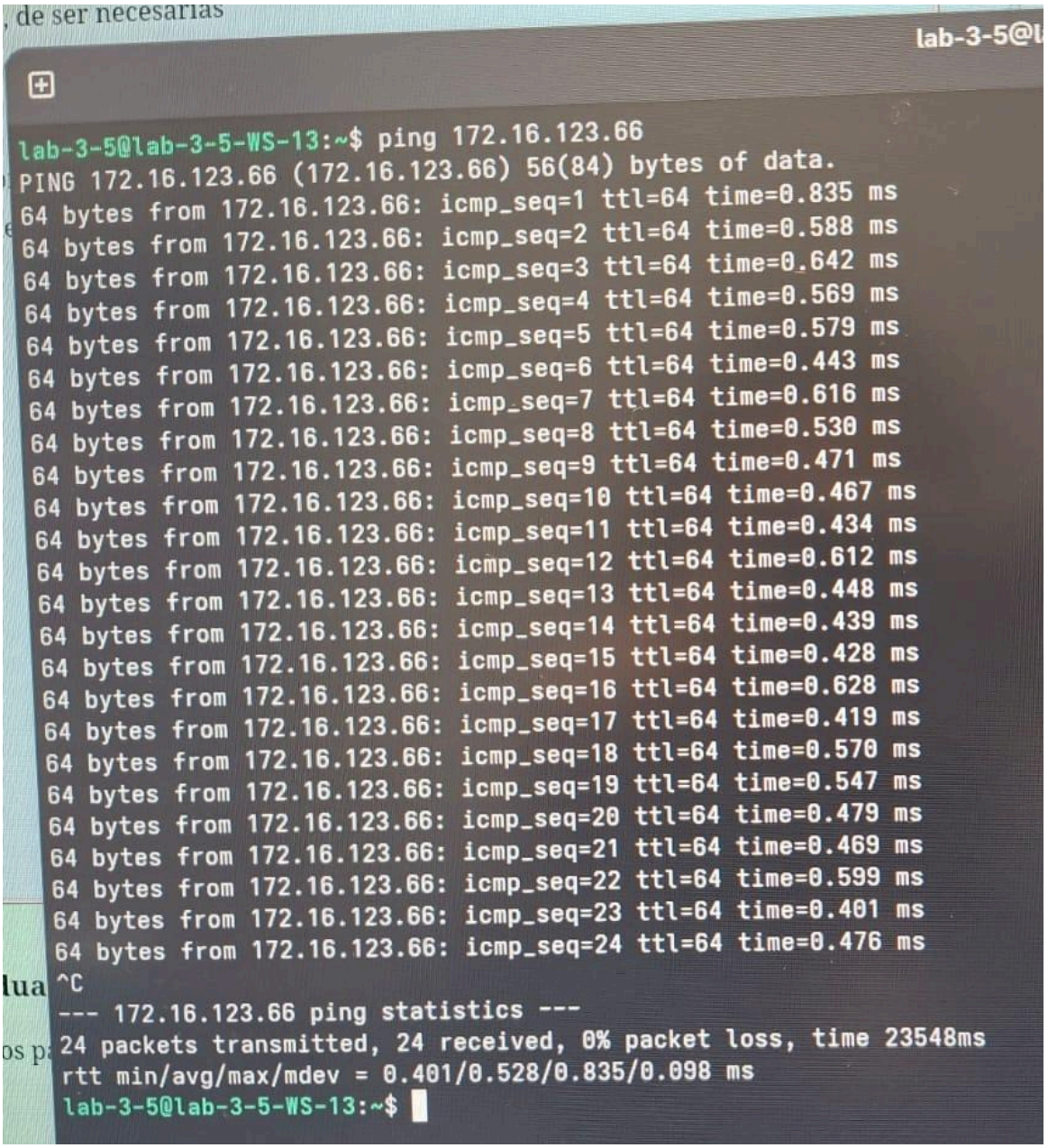
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/18
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/19
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/20
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/21
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/22
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/23
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast
--More-- !
--More-- interface FastEthernet0/24
--More-- switchport access vlan 230
--More-- switchport mode access
--More-- spanning-tree portfast

--More-- !
--More-- interface GigabitEthernet0/1
--More-- switchport trunk allowed vlan 230
--More-- switchport mode trunk
--More-- !
--More-- interface GigabitEthernet0/2
--More-- !
--More-- interface Vlan1
--More-- no ip address
--More-- !
--More-- interface Vlan230
--More-- ip address 172.16.123.50 255.255.255.240
--More-- !
--More-- ip default-gateway 172.16.123.49
--More-- ip http server
--More-- ip http secure-server
--More-- !
--More-- line con 0
--More-- line vty 5 15
--More-- !
--More-- end

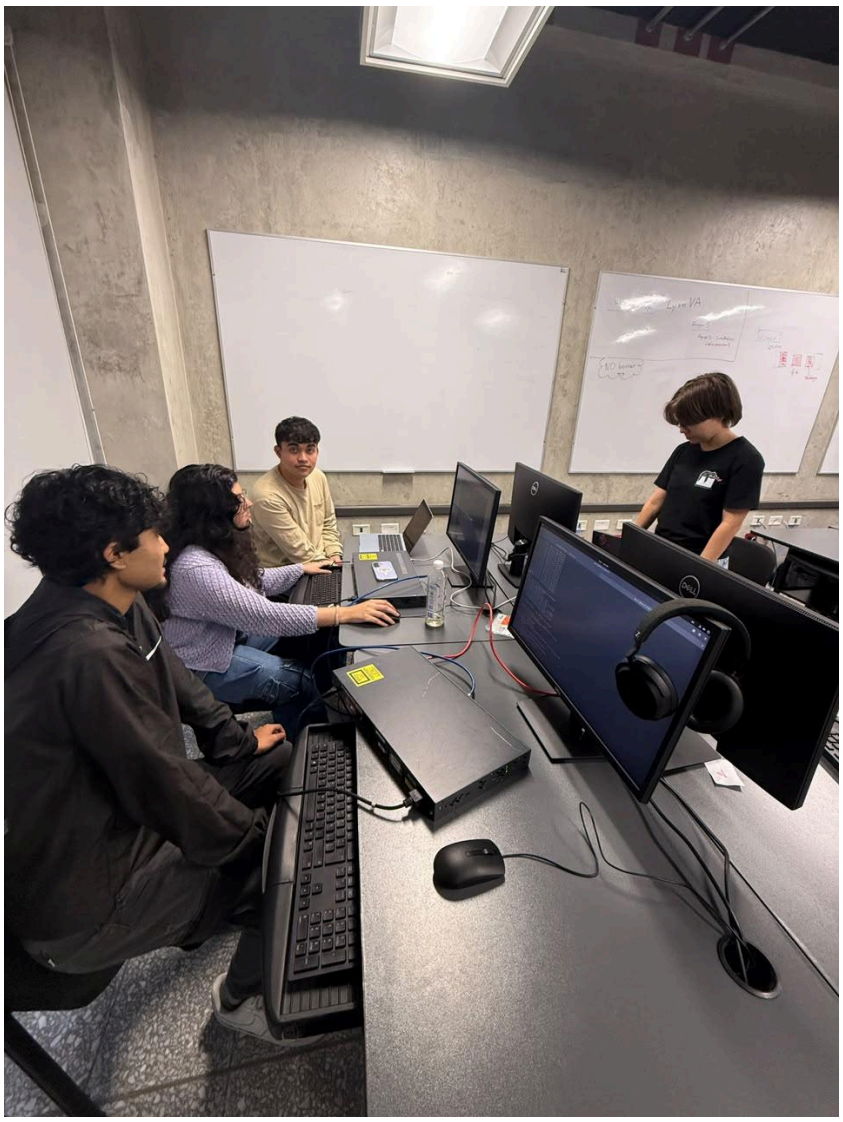
4.Evidencia de PING entre switches

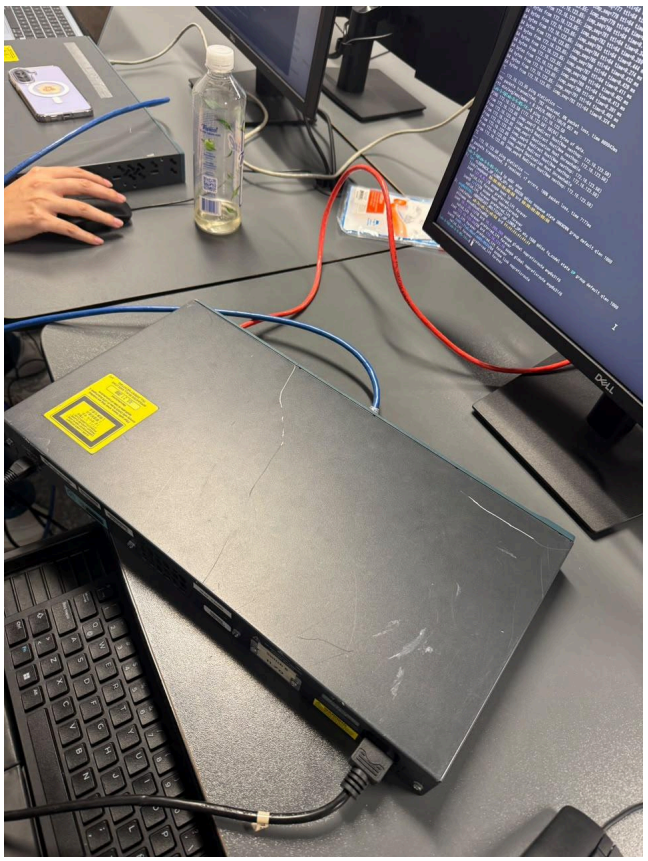


L3 a L2 los 2



Ping L2 isla 3 a L2 isla 4




DELL

